

ANIL PRADHAN

Full Stack Developer · AI/Agent Systems · Backend Engineering

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EDUCATION

B.Tech – CSE | **CGPA: 9.11/10** | ITER, SOA University, Bhubaneswar 2024 – 2028
Class XII – PCM with IT | **85%** | Odisha Board of Secondary Education 2022 – 2024

EXPERIENCE

Campus Ambassador | E-Cell, IIT Kanpur · Remote Dec 2025 – Jan 2026
► Represented IIT Kanpur's E-Summit at ITER, SOA University; boosted student participation via tech talks and networking sessions.

PROJECTS

AI Career Mentor Feb 2026 – Present
Next.js 14 · FastAPI · LangGraph · ChromaDB · Groq · NVIDIA NIM · Gemini 2.5 Flash · Neon Postgres · Upstash Redis · Google OAuth · Serper · Tavily · Docker
► Architected a production-grade career coaching platform using a LangGraph static DAG orchestrating 5 parallel AI agents (Resume Analyst → Market Researcher + LinkedIn Optimizer → Roadmap Builder) with Pydantic validation/repair loops and circuit-breaker fallbacks — delivering complete career diagnostics in under 60 seconds across 254+ commits.
► Built a Hybrid Semantic RAG Engine using ChromaDB persistent vector store (auto-seeded at startup with 50+ curated resources) with in-memory MockRAG fallback preventing OOM crashes on Render's 512MB RAM containers — enriching personalized 8-week roadmaps with verified YouTube, GitHub, and documentation resources via multi-source deduplication pipeline.
► Engineered real-time WebSocket mock interview system via OpenAI SDK over Groq/NVIDIA — sub-2s token streaming, 7-question adaptive state machine with 3 difficulty tiers (FAANG=Hard, Product=Medium, Service=Easy), Edge-TTS voice synthesis, and crash-resilient WebSocket cleanup preventing memory leaks across concurrent sessions.
► Implemented Dynamic Multi-Provider LLM routing across Groq (Llama 3.3 70B), NVIDIA NIM (Llama 3.3 70B-Instruct), and Gemini 2.5 Flash with automatic fallback chains (groq→google, nvidia→google), deterministic fallback engines for malformed outputs, and circuit-breaker mechanisms ensuring 100% system availability under free-tier API rate limits.
► Hardened market intelligence pipeline integrating Serper.dev, Tavily AI, and DuckDuckGo with regex-based JSON extraction, async link validation (10 concurrent workers), and resource scoring engine (domain authority weights + freshness signals) — delivering real-time salary benchmarks and hiring volumes with zero static mock data.
► Designed secure auth pipeline (Google OAuth 2.0, JWT + Refresh Token rotation preventing replay attacks), SHA-256 Redis caching (7-day TTL), SlowAPI 100 req/hr rate limiting, PostgreSQL connection pooling (pool_size=3, pool_pre_ping for Neon compatibility), deployed via Docker + GHCR + GitHub Actions CI/CD (ESLint + 83 pytest tests + pip-audit) across Vercel + Render + Neon Postgres. Submitted to Microsoft AI DevDays (\$80K+) and Amazon Nova AI (\$95K+) hackathons.

VolunteerIQ Mar 2026 – Apr 2026
Next.js 14 · FastAPI · Firebase Auth · Gemini 2.5 Flash · Mapbox GL · react-map-gl · PostgreSQL · Docker
► Built an AI-powered NGO coordination platform for Google Solution Challenge 2026, integrating Gemini 2.5 Flash to process multi-format field surveys (PDF, CSV, DOCX, TXT) and extract urgent community problems with urgency scores and auto-generated deployable field tasks.
► Implemented AI volunteer matching system ranking candidates by skill match, location proximity, and availability with a 0–100 score, alongside a context-aware AI chat assistant querying live PostgreSQL data to answer plain-English NGO operational queries.
► Deployed a Mapbox GL live operations map with geocoding cache, Firebase JWT authentication verified server-side on every protected request, and Docker + docker-compose containerization with Gemini-to-Groq automatic fallback under rate limits.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · Java
Frontend: Next.js 14 · React · Lucide React · react-map-gl
Backend: FastAPI · LangGraph · OpenAI SDK (via Groq/NVIDIA) · SQLAlchemy · Alembic · Pydantic · Uvicorn · pdfplumber · edge-tts · WebSocket · SlowAPI
AI / LLM & RAG: Groq (Llama 3.3 70B) · NVIDIA NIM · Gemini 2.5 Flash · ChromaDB · Serper.dev · Tavily AI
Databases: Neon Postgres · SQLite · Upstash Redis
Auth & Security: Google OAuth 2.0 · Firebase Authentication · JWT · bcrypt · google-auth
DevOps & Infra: Docker · GitHub Actions · Vercel · Render.com · GHCR · CI/CD
Tools: VS Code · Git · Swagger UI · Postman · Mapbox GL · Cline · GitHub Copilot · Gemini Code Assist · OpenAI Codex · AntiGravity

ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024–25): Completed with 75 Arcade Points; learned multiple cloud skills across GCP services.
- GeeksForGeeks – Java DSA: 80+ day continuous streak; 200+ problems solved across GFG and LeetCode (Arrays, Strings, Recursion, DP, Graphs). Certificate + T-shirt awarded.
- Active Tech Community: Attending multiple offline tech events, building professional network, and connecting with developers at GDG sessions and bootcamps.
- Top Tech Hackathons: Shipped projects in 3 major hackathons — Microsoft AI Dev Days (\$80K+), Amazon Nova AI (\$95K+), and Google Solution Challenge 2026. Consistently building and competing.